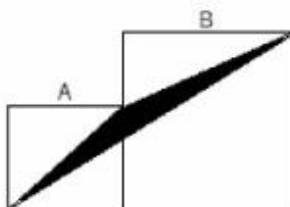


JMO – 2022 (11th DEC-2022)

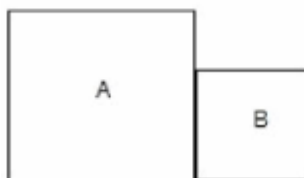
Answer all questions. All questions carry equal marks.

Full Marks 100

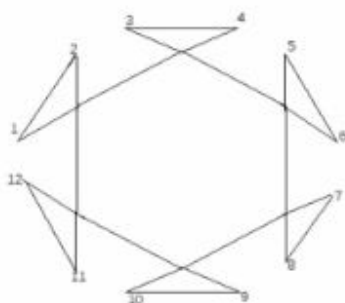
- Given that in the figure, A and B are two squares of sides 3 cm and 8 cm respectively. Find the area of shaded portion.



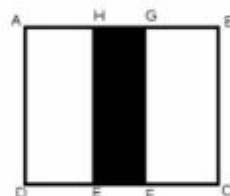
- 2 persons A and B have together 100 balls. $\frac{1}{3}$ of the total number of balls of A is 16 more than $\frac{1}{10}$ of the number of balls of B. How many balls each of A and B have?
- In the given figure A and B are two squares. Sum of the areas of these two squares is 2020 sq cm. Find the perimeter of this given figure.



- How many digits are there in the product $4^{11} \times 5^{22} \times 10^{44}$
- Evaluate $\frac{1}{1 \times 2} + \frac{1}{2 \times 3} + \frac{1}{3 \times 4} + \dots$ upto 100 terms.
- In the given figure:
 $\angle 1 + \angle 2 + \angle 3 + \angle 4 + \angle 5 + \angle 6 + \angle 7 + \angle 8 + \angle 9 + \angle 10 + \angle 11 + \angle 12 = ?$



- In the given figure ABCD is a square of side 8 cm, Given $AG = EC = 5$ cm, find the area of shaded portion.



- Find the sum of first 2022 terms in the following: 1, 2, 3, 4; 1, 2, 3, 4; 1, 2, 3, 4;
- Find the sum of all digits of the result in the subtraction: $100^{999} - 999$
- Write in increasing order of the following numbers 2^{222} , 22^{22} , 222^2 , 2222

JMO QUESTIONS (ORISSA)

11. If m person do a job in d days, then find the number of days in which $(m+r)$ persons can do the same job.
12. What will be the remainder if 2^{2022} is divisible by 2022.
13. If an article is sold with a discount of 5% then the profit is 5%. What will be the percentage of profit if no discount is given?
14. One sheet of a note with 32 pages is torn. If the sum of pages of remaining sheets is 495, then what is the page numbers of the two pages of torn sheet?
15. Find the sum of the following : $1^2 - 2^2 + 3^2 - 4^2 + \dots - 98^2 + 99^2$
16. How many zeroes are there at the end in the following product?
17. The hour hand was between 4 and 5, the minute hand was in between 5 and 6 when I left home. When I returned home I observed that the location of hour and minute hand are interchanged. At what time did I leave home?
18. Two buses left Cuttack and Balasore at the same time and moved towards Balasore and Cuttack respectively. They reached their destination respectively after one hour and four hours of their meeting time. If the first bus has speed 60 km/hour, what is the speed of the second bus?
